

RW4 Sheet Module - ProducerUA

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Input workbook: full_uah.xlsx

Source sheet: Producer_UA

Rows (clean): 10758

Date range: 2021-02-01 .. 2025-12-29

Products: 5

Stats rows: 15

Tests rows: 15

Interpretation option: use tests table -> recommended_action/recommended_model_family per product.

Table: clean

	date	raw_product	product	standardized_type	brand	region	price	price_real_input	price_linear_input	price_pchip_input	unit_ok	subcategory	unit_family	liter_equiv_allowed	maturation_type	mapping_quality_flag	matched_pattern	comparison_family	price_variant	reconstruction_variant	observed_price	baseline_price	fat_band	pack_band	brand_normalized	segment_key	shock_dummy	admissible_for_level_model	admissibility_reason
ProducerUA	2021-02-01	Кефір	Кефір	milk		Україна	5.156141895433997	nan	5.156141895433997	5.156141895433997	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.156141895433997	5.156141895433997	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-02	Кефір	Кефір	milk		Україна	5.151731428728376	nan	5.151731428728376	5.146901289333981	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.151731428728376	5.151731428728376	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-03	Кефір	Кефір	milk		Україна	5.147320962022755	nan	5.147320962022755	5.13956232396571	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.147320962022755	5.147320962022755	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-04	Кефір	Кефір	milk		Україна	5.142910495317135	nan	5.142910495317135	5.133950042769015	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.142910495317135	5.142910495317135	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-05	Кефір	Кефір	milk		Україна	5.138500028611515	nan	5.138500028611515	5.129889489183729	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.138500028611515	5.138500028611515	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-06	Кефір	Кефір	milk		Україна	5.134089561905894	nan	5.134089561905894	5.127205706649685	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.134089561905894	5.134089561905894	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-07	Кефір	Кефір	milk		Україна	5.129679095200273	nan	5.129679095200273	5.125723738606716	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.129679095200273	5.129679095200273	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-08	Кефір	Кефір	milk		Україна	5.125268628494652	nan	5.125268628494652	5.125268628494652	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.125268628494652	5.125268628494652	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-09	Кефір	Кефір	milk		Україна	5.132536719016345	nan	5.132536719016345	5.126929590310015	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.132536719016345	5.132536719016345	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-10	Кефір	Кефір	milk		Україна	5.139804809538039	nan	5.139804809538039	5.131497366976496	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.139804809538039	5.139804809538039	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-11	Кефір	Кефір	milk		Україна	5.147072900059732	nan	5.147072900059732	5.138349295324693	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.147072900059732	5.147072900059732	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-12	Кефір	Кефір	milk		Україна	5.154340990581426	nan	5.154340990581426	5.146862712185197	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.154340990581426	5.154340990581426	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-13	Кефір	Кефір	milk		Україна	5.161609081103119	nan	5.161609081103119	5.156414954388605	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.161609081103119	5.161609081103119	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-14	Кефір	Кефір	milk		Україна	5.168877171624813	nan	5.168877171624813	5.166383358765509	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.168877171624813	5.168877171624813	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-15	Кефір	Кефір	milk		Україна	5.176145262146506	nan	5.176145262146506	5.176145262146506	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.176145262146506	5.176145262146506	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-16	Кефір	Кефір	milk		Україна	5.189653683829126	nan	5.189653683829126	5.188326885773577	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.189653683829126	5.189653683829126	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-17	Кефір	Кефір	milk		Україна	5.203162105511746	nan	5.203162105511746	5.204535621311408	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.203162105511746	5.203162105511746	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-18	Кефір	Кефір	milk		Україна	5.216670527194365	nan	5.216670527194365	5.222620561224653	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.216670527194365	5.216670527194365	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-19	Кефір	Кефір	milk		Україна	5.230178948876985	nan	5.230178948876985	5.240430797977967	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.230178948876985	5.230178948876985	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-20	Кефір	Кефір	milk		Україна	5.243687370559605	nan	5.243687370559605	5.255815424036	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.243687370559605	5.243687370559605	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-21	Кефір	Кефір	milk		Україна	5.257195792242225	nan	5.257195792242225	5.266623531863409	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.257195792242225	5.257195792242225	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-22	Кефір	Кефір	milk		Україна	5.270704213924844	nan	5.270704213924844	5.270704213924844	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.270704213924844	5.270704213924844	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-23	Кефір	Кефір	milk		Україна	5.112765007224075	nan	5.112765007224075	5.209462480714342	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	5.112765007224075	5.112765007224075	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-24	Кефір	Кефір	milk		Україна	4.954825800523306	nan	4.954825800523306	5.051523274013573	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	4.954825800523306	4.954825800523306	unknown	unknown	milk fermented_milk unkno...	0	1	uah_per_kg	
ProducerUA	2021-02-25	Кефір	Кефір	milk		Україна	4.796886593822537	nan	4.796886593822537	4.835565583218644	1	fermented_milk	liquid	1	fresh	matched	кефір	liquid_dairy	observed	observed	4.796886593822537	4.796886593822537	unknown	unknown					

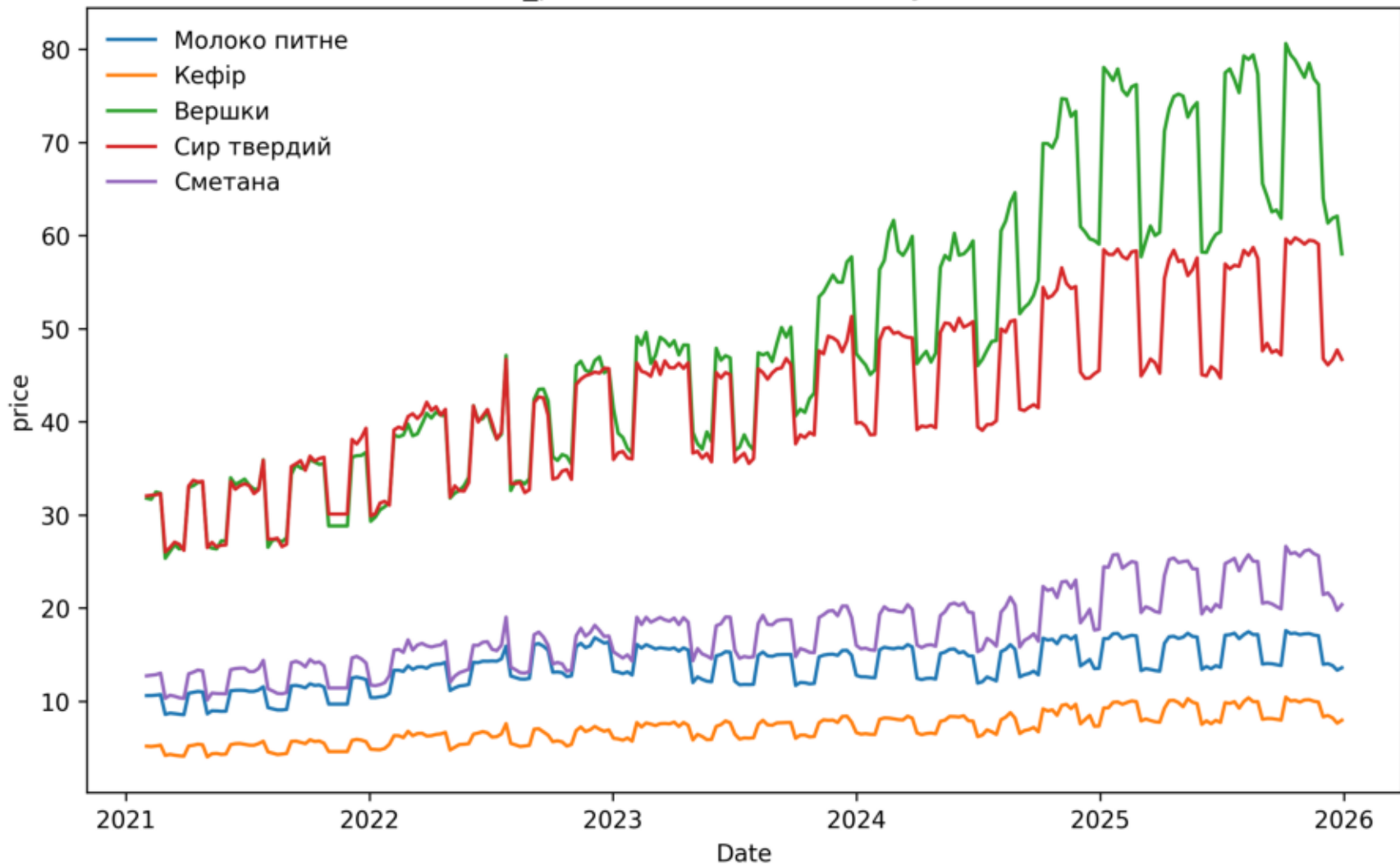
Table: descriptive_stats

source	product	standardized_type	series_variant	count	missing	imputed_share	mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30
ProducerUA	Вершки	cream	linear	1793	0	0.0	48.96868060267151	46.74153573049654	15.148986812878368	25.33829767439181	80.6094690835544	27.600185383489947	36.36707797078087	59.74098426601655	77.22580444852069	0.47336017870275315	-0.8545016563566912	0.3093607307045129	0.005322324695203865	0.011421651173579498
ProducerUA	Вершки	cream	pchip	1793	0	0.0	48.97055213864296	46.72973481286407	15.181701623626628	25.33829767439181	80.6094690835544	27.40406478831619	36.36492641358905	59.72377660793099	77.18480902003982	0.4751414743712641	-0.8485109964322266	0.31001695836806087	0.0058263033694525455	0.012334244380025217
ProducerUA	Вершки	cream	real	1793	0	nan	48.96868060267151	46.74153573049654	15.148986812878368	25.33829767439181	80.6094690835544	27.600185383489947	36.36707797078087	59.74098426601655	77.22580444852069	0.47336017870275315	-0.8545016563566912	0.3093607307045129	0.005322324695203865	0.011421651173579498
ProducerUA	Кефір	milk	linear	1793	0	0.0	7.028310370107854	6.876646514564298	1.5996261198885564	4.012087564132884	10.4258708986717	4.487797121394208	5.781080729982296	8.061562194605138	9.940093163754016	0.22974970008080567	-0.7512064619862322	0.22759753563131407	0.006102622143983761	0.012254310067779804
ProducerUA	Кефір	milk	pchip	1793	0	0.0	7.0280435593766395	6.868252196654961	1.6071247396071509	4.012087564132884	10.4258708986717	4.464451500620355	5.778258076123241	8.054167466536526	9.944100450728431	0.23280204649450492	-0.749407687707238	0.22867313300341818	0.0065886417868524245	0.013164841461109012
ProducerUA	Кефір	milk	real	1793	0	nan	7.028310370107854	6.876646514564298	1.5996261198885564	4.012087564132884	10.4258708986717	4.487797121394208	5.781080729982296	8.061562194605138	9.940093163754016	0.22974970008080567	-0.7512064619862322	0.22759753563131407	0.006102622143983761	0.012254310067779804
ProducerUA	Молоко питне	milk	linear	1793	0	0.0	13.730708268001278	13.814370998998143	2.2993826989979382	8.54271736699659	17.574624928800496	9.25612986048701	12.113219396565553	15.638478361839177	17.122309273610746	-0.3210835044458281	-0.7227439011128887	0.16746278881742274	0.0049668884737525485	0.010809505073854649
ProducerUA	Молоко питне	milk	pchip	1793	0	0.0	13.730497012744845	13.778628913288006	2.3130366110564378	8.54271736699659	17.574624928800496	9.224816406718139	12.084963650088689	15.663908931897963	17.133016708295635	-0.3125118139109012	-0.7377180700466197	0.16845978764712188	0.005464850645559347	0.011677347965764778
ProducerUA	Молоко питне	milk	real	1793	0	nan	13.730708268001278	13.814370998998143	2.2993826989979382	8.54271736699659	17.574624928800496	9.25612986048701	12.113219396565553	15.638478361839177	17.122309273610746	-0.3210835044458281	-0.7227439011128887	0.16746278881742274	0.0049668884737525485	0.010809505073854649
ProducerUA	Сир твердий	hard_cheese	linear	1793	0	0.0	42.729036041396675	42.38203210401479	8.711941430655303	26.01486142306449	59.75215453866704	27.646081054578254	36.00774888688341	48.52832059070337	58.11728139013326	0.13180462877346638	-0.7535363727679076	0.2038880873000508	0.005467038287097556	0.011731971519494582
ProducerUA	Сир твердий	hard_cheese	pchip	1793	0	0.0	42.72971586454939	42.26148012864311	8.750381135714294	26.01486142306449	59.75215453866704	27.46401788348839	36.00150486533281	48.46083815683446	58.18263338448224	0.13354113120160763	-0.7564203619452745	0.20478444470476873	0.006005566622862169	0.012691614912667373
ProducerUA	Сир твердий	hard_cheese	real	1793	0	nan	42.729036041396675	42.38203210401479	8.711941430655303	26.01486142306449	59.75215453866704	27.646081054578254	36.00774888688341	48.52832059070337	58.11728139013326	0.13180462877346638	-0.7535363727679076	0.2038880873000508	0.005467038287097556	0.011731971519494582
ProducerUA	Сметана	sour_cream	linear	1793	0	0.0	17.470030270593426	16.97507948765535	4.0729931256160175	10.12501466319408	26.61583297980757	11.216217504302845	14.37866541837231	19.95342012265521	25.14485263517279	0.36950655961754664	-0.5955186676790101	0.233141732586	0.005729570178683589	0.011824939482185916
ProducerUA	Сметана	sour_cream	pchip	1793	0	0.0	17.469559665540558	16.96142290035107	4.089021987144437	10.12501466319408	26.61583297980757	11.168474685614454	14.39528686818635	19.96135661470661	25.190267447028777	0.3700902007640757	-0.5962643066040694	0.23406554403372887	0.006226182242742337	0.01273092749920205
ProducerUA	Сметана	sour_cream	real	1793	0	nan	17.470030270593426	16.97507948765535	4.0729931256160175	10.12501466319408	26.61583297980757	11.216217504302845	14.37866541837231	19.95342012265521	25.14485263517279	0.36950655961754664	-0.5955186676790101	0.233141732586	0.005729570178683589	0.011824939482185916

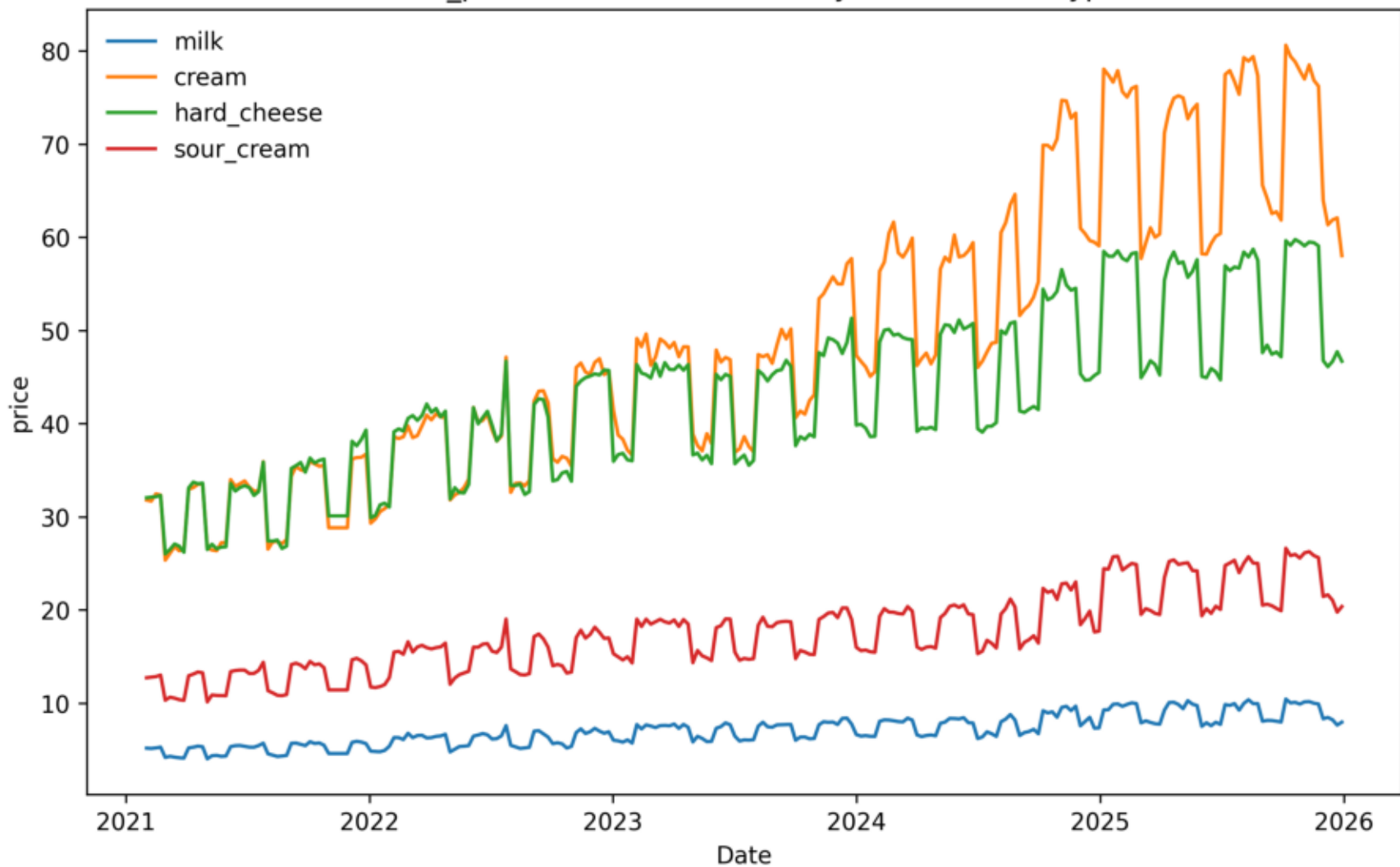
Table: tests

source	product	standardized_type	series_variant	n_obs	adf_p	kpss_p	ljungbox_p	bp_p	white_p	jb_p	stability_flag	stability_drift	action_label	integration_class	ac_risk	het_risk	non_normal_risk	stability_risk_class	recommended_action	recommended_model_family	avoid	diag_note
ProducerUA	Вершки	cream	linear	257	0.8658129740633709	0.01	4.796087442050229e-31	0.8055676604875804	0.0005501022648871136	1.2922206264232943e-10	1	23.833973336649805	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Вершки	cream	pchip	257	0.8760270267902056	0.01	3.744325435099192e-30	0.8058355991805275	0.0005412382870159002	2.66433216374388e-09	1	23.838851005298253	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Вершки	cream	real	257	0.8658129740633709	0.01	4.796087442050229e-31	0.8055676604875804	0.0005501022648871136	1.2922206264232943e-10	1	23.833973336649805	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Кефір	milk	linear	257	0.8082564821149362	0.01	4.426817849036492e-24	0.7747510637802231	0.00036812593420594884	8.936344375963147e-08	1	2.324318559842668	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Кефір	milk	pchip	257	0.800070528039281	0.01	5.160492590491831e-22	0.7742756624362586	0.0008426461575638194	1.793401290886123e-05	1	2.3238799403809054	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Кефір	milk	real	257	0.8082564821149362	0.01	4.426817849036492e-24	0.7747510637802231	0.00036812593420594884	8.936344375963147e-08	1	2.324318559842668	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Молоко питне	milk	linear	257	0.21026477235797558	0.01	5.6949823356599925e-34	0.8186485207887365	3.9129119230073645e-06	2.3659252678538565e-12	1	2.286466827819794	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Молоко питне	milk	pchip	257	0.21055856580296928	0.01	7.787769095459672e-33	0.8460215784457581	5.186227577685155e-06	1.2730292677746694e-10	1	2.286024136345336	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Молоко питне	milk	real	257	0.21026477235797558	0.01	5.6949823356599925e-34	0.8186485207887365	3.9129119230073645e-06	2.3659252678538565e-12	1	2.286466827819794	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Сир твердий	hard_cheese	linear	257	0.6930796464906741	0.01	5.500918266223269e-32	0.80562401373873	0.00012318158117180787	1.7445756486849289e-09	1	12.198741998956386	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Сир твердий	hard_cheese	pchip	257	0.6967124107766377	0.01	3.3824610279866177e-31	0.7827693989340003	0.0001453123988218523	1.658707798176277e-08	1	12.200580080282919	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Сир твердий	hard_cheese	real	257	0.6930796464906741	0.01	5.500918266223269e-32	0.80562401373873	0.00012318158117180787	1.7445756486849289e-09	1	12.198741998956386	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Сметана	sour_cream	linear	257	0.9108841922545056	0.01	5.606735121404709e-28	0.7416238029377913	0.0010537779476076968	7.725824662010651e-13	1	5.7630888906094865	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Сметана	sour_cream	pchip	257	0.904867175942224	0.01	2.7316314079275357e-26	0.7626731761750097	0.0018807518946076598	5.707929493580508e-10	1	5.762069489456948	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ProducerUA	Сметана	sour_cream	real	257	0.9108841922545056	0.01	5.606735121404709e-28	0.7416238029377913	0.0010537779476076968	7.725824662010651e-13	1	5.7630888906094865	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...

sheet_producerua: Time Series by Product



sheet_producerua: Time Series by Standardized Type



sheet_producera_distribution.png

sheet_producera: Distribution of price

